Tutorial Topic 2
Limits & Continuity

(1) Find $\lim_{x\to 1} f(x)$ where $f(x)=\begin{cases} \frac{1}{x+2} & , \quad x<1 \\ 1-2x & , \quad x>1 \end{cases}$.

(2) Draw the graph of the function $k(x)=\begin{cases} -1 & , \quad x<0 \\ 0 & , \quad x=0 \\ 1 & , \quad x>0 \end{cases}$.

Then, evaluate:

(a) $\lim_{x\to 0^-} k(x)$ (b) $\lim_{x\to 0^+} k(x)$ (c) $\lim_{x\to 0} k(x)$ (d) $\lim_{x\to 0^-} |k(x)|$

(3) Evaluate the following limits if it exists. If it does not exist, explain why.

(a) $\lim_{x\to 2}\left(3x^3-4x^2+2\right)$

(b) $\lim_{x\to -2}\left(5x^3-2x+3\right)$

(c) $\lim_{x\to -3}\dfrac{x^2-4x+1}{3x-4}$

(d) $\lim_{x\to 0}\dfrac{x^2-4x+1}{3x-4}$

(e) $\lim_{x\to 5}\left(2x-\dfrac{6}{x}\right)$

(f) $\lim_{h\to 0}\dfrac{6+\sqrt{h^2-3h+4}}{2-h}$

(g) $\lim_{x\to 2}\dfrac{\sqrt{5x^3+3x+3}}{x^2-5}$

(h) $\lim_{x\to -1}\dfrac{x^2-1}{x+1}$

(i) $\lim_{x\to +\infty}\dfrac{3x+1}{2x-5}$

(j) $\lim_{x\to +\infty}\dfrac{(x-4)(x-5)}{(x+1)(x-7)}$

(4) Given f(x) = $\begin{cases} x+2 & , \quad x<-2 \\ x^2-4 & , \quad x=-2 \\ 2x+4 & , \quad x>-2 \end{cases}$, is f(x) continuous at x = -2 ?

(5) Discuss the continuity of the following functions :

(a) $f(x)=\dfrac{1}{x}$ (b) $g(x)=\dfrac{x^2-1}{x+1}$ (c) $h(x)=\begin{cases} x+1, & x<1 \\ 2-x, & x\geq 1 \end{cases}$

(6) For what values of the constant A is the function

$$f(x)=\begin{cases} Ax+5, & x<1 \\ x^2-3x+4, & x\geq 1 \end{cases}$$

continuous for all real x ?